



**Time:** 5 minutes

**Outcome:** Apply Profit-Maximizing Output & Monopoly Price.

**Difficulty:** ●●○ Intermediate

## Profit-Maximizing Output & Monopoly Price



### THE CORE IDEA

A single-price monopolist chooses output where marginal revenue (MR) crosses marginal cost (MC) from above, so extra units change from adding to reducing profit. It then reads price from demand at that output. This interior rule must be checked against shutdown;  $MR = MC$  alone does not guarantee positive profit.



### HOW TO RECOGNIZE IT

- Find the output where MR crosses MC from above in the interior solution.
- Move vertically from that output to the demand curve to find the monopoly price.
- Keep the output decision and price decision separate: marginal conditions determine quantity, demand determines price.
- Do not set price equal to marginal revenue or marginal cost unless the graph happens to imply it.

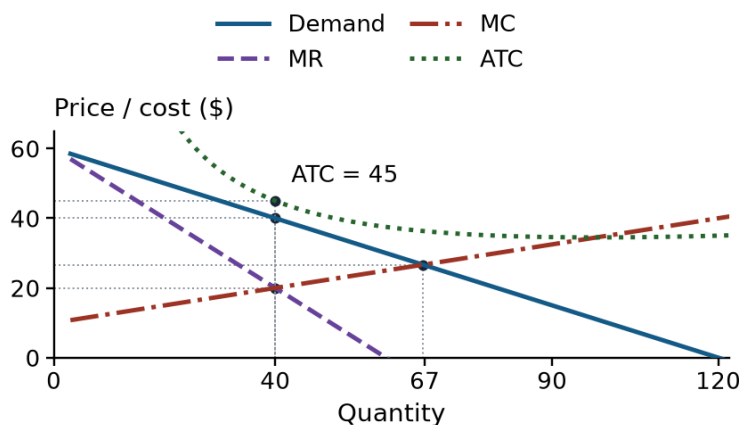


### WATCH OUT

Read quantity where MR crosses MC, then price on demand. An upward-sloping MC curve is not essential; the relevant condition is MR crossing MC from above.



### WORKED EXAMPLE: FINDING MONOPOLY PRICE



The monopolist first finds output where  $MR = MC$ , which occurs at 40 units and a marginal value of \$20. It then moves up to the demand curve, where buyers will pay \$40 for 40 units. The profit-maximizing choice is therefore  $Q = 40$  and  $P = \$40$ , not the \$20 MR value.



### CHECK YOURSELF

What is the correct order for solving a monopoly graph?



**READY?**

Return to the game and master this concept.